Test 9 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"Leila participated in a science exhibition at her school. She created a project demonstrating how solar energy can be used to power small devices. She explained the benefits of renewable energy and how it helps reduce pollution. Many students and teachers appreciated her work, and she received a certificate for innovation and creativity."

a) What type of project did Leila present?
b) Name two points Leila explained about her project.
c) How did the school community respond to her work?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Join the sentences using a suitable conjunction:
"He forgot his notebook. He could not complete the homework."

b) Identify all verbs in the sentence:
"The students enthusiastically participated in the annual sports competition."

c) Correct the grammatical error:
"Each of the teachers have prepared their lessons carefully."

d) Rewrite in indirect speech:
"Ahmed said, 'I will submit my assignment before the deadline.'"

3. Writing
Write a short essay (120–150 words) on the topic: "The Advantages and Challenges of Online Learning". Include benefits, limitations, and personal opinion.

Section B: Mathematics

1. Algebra
a) Solve for x:
$6x + 15 = 3x + 24$

b) Factorize:
$x^2 – 8x + 15$

c) Simplify:
$4a^2b – 5ab^2 + 3a^2b$

2. Geometry & Mensuration
a) A trapezium has parallel sides 12 cm and 8 cm, with height 10 cm. Find its area.
b) A cone has a radius of 7 cm and height 24 cm. Find its volume.

3. Ratio, Proportion & Percentage
a) A bicycle costs $360. It is sold at a 20% discount. Find the selling price.
b) If 10 notebooks cost $50, find the cost of 35 notebooks.

4. Data Handling
Marks of 12 students in a science quiz: 14, 16, 15, 17, 18, 14, 16, 15, 19, 17, 16, 18.
a) Calculate the mean, median, and mode.
b) Present the data in a simple frequency table.

Section C: Science

1. Biology
a) What are the functions of the human heart?
b) Explain respiration in plants with the chemical equation.
c) List two differences between fish and amphibians.

2. Physics
a) A bike accelerates from 5 m/s to 25 m/s in 4 seconds. Calculate acceleration.
b) Define mechanical energy and give one example.
c) Distinguish between reflection and refraction of light.

3. Chemistry
a) Identify whether the following are elements, compounds, or mixtures:

- **Hydrogen**
- **Carbon dioxide**
- **Saltwater**

b) Write a balanced chemical equation for the reaction between iron and hydrochloric acid.

4. Environmental Science
a) Name two causes of soil pollution and their effects.
b) Suggest three methods to reduce electricity consumption in schools.

Section D: Social Studies

1. History
a) Name two major revolutions of the 19th century and their outcomes.
b) Explain the importance of the Renaissance in European history.
c) Identify the Axis Powers during World War II.

2. Geography
a) Define climate and describe two factors that affect it.
b) Name the highest mountain in Africa and its location.
c) Explain two consequences of urbanization on the environment.

3. Civics / Political Science
a) Compare democratic and authoritarian governments.
b) State two rights and two responsibilities of citizens in a democracy.

4. Economics
a) Explain the concept of inflation with an example.
b) Differentiate between fixed and variable costs with examples.

Section E: ICT / Computer Studies

1. Computer Basics
a) Define software and give one example.
b) Name one input device and one primary storage device.

2. Applications
a) Which spreadsheet function counts the number of entries in a dataset?
b) Mention two uses of word processing software in schools.

3. Programming Basics
a) Explain the concept of a function in programming with an example.
b) Write a Python code snippet to print numbers from 10 to 1 in reverse order.

4. Cyber Awareness
a) Define ransomware and suggest one way to prevent it.
b) Mention two steps to protect personal information online.

Section F: Islamiyat

1. Quran & Hadith
a) Mention two lessons taught by the Quran about kindness.
b) Narrate a Hadith that encourages charity.

2. Aqidah / Belief
a) List the five pillars of Islam.
b) Explain the importance of believing in Qadr (divine predestination).

3. Fiqh / Worship
a) Explain the significance of performing Salah on time.
b) Describe the spiritual benefits of fasting in Ramadan.

4. Islamic Ethics & Morality
a) How does Islam encourage fair trade in markets?
b) Explain the importance of protecting animals according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain the use of contrast in artwork.
b) Name two secondary colors.

2. Music
a) Define tempo and dynamics in music.
b) Give an example of a wind instrument from any culture.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises that improve cardiovascular health.
b) List one indoor game and one outdoor game.

2. Health & Hygiene
a) Why is drinking enough water important for teenagers?
b) Mention two ways to maintain oral hygiene.

3. Safety & Environment
a) State one safety precaution when using electrical appliances.